Unsupervised Deep Learning of ECGs

Master Thesis - Software Design

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01 Introduction & RQs

Developing a Convolutional Neural Network (CNN) based autoencoder to explore the latent space of 1ECGs, aiming to discover subtle cardiac abnormalities that traditional clinical rules miss.

Do specific segments of the latent space clusters align with established diagnostic categories or prognosis?

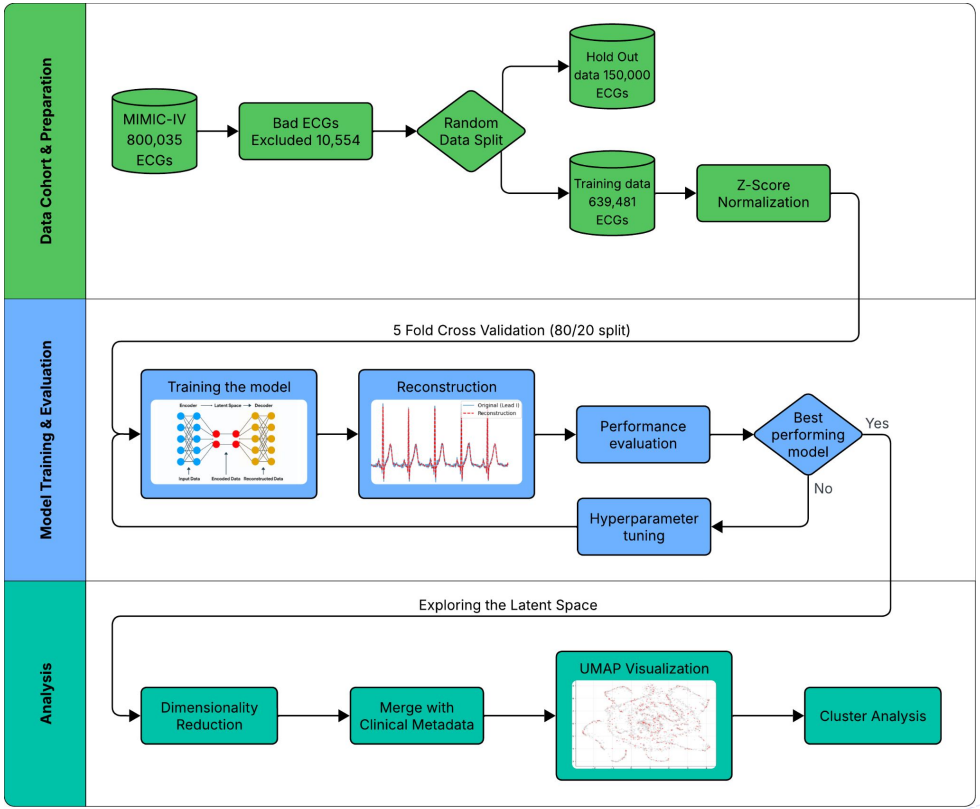
To what extent can unsupervised deep learning capture ECG abnormalities?

02 The Dataset

MIMIC-IV-ECG Dataset from PhysioNet

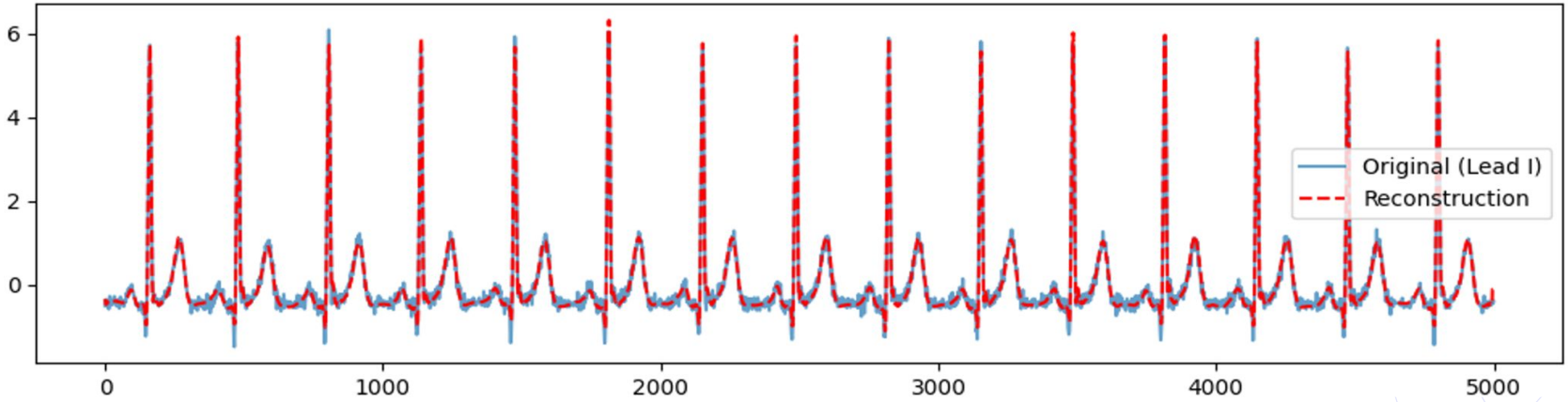
- 160,000 Unique patients
- 800,035 ECGs
  - 8 Leads
  - 10 Seconds in length
  - 500 Hz
  - Resulting 8 columns and 5,000 rows for each ECG
- Beth Israel Deaconess Medical Center

03 Methodology Overview

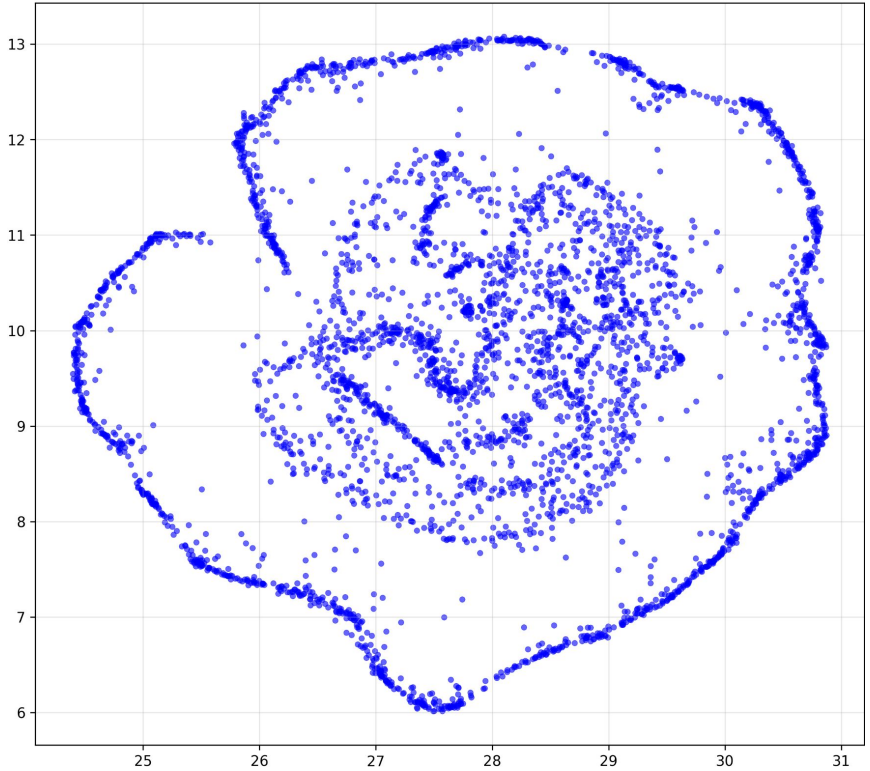


04 Results Overview

Average Validation MSE	Average Validation RMSE	Confidence Interval RMSE	Average Validation MAE	Average Validation R^2
0.047	0.217	[0.213, 0.220]	0.133	0.953



04 Results - Latent Space



05 Existing Research

Category	My Model	Eravci (Wavelet)	Kim (LSTM)	Song (CREMA)	Silva (Sparse AE)	Jiang (CAE-LSTM)	Shan (ECG-AAE)	Zhang (LRA-AE)	Friedman (DAE)
Main Goal	Explore latent space for anomaly detection	Automated Atrial Fibrillation detection	AFib detection using P-wave segments	General-purpose foundation model	Beat-by-beat mobile AFib detection	Predict future paroxysmal AFib attacks	Semi-supervised arrhythmia detection	Filter noise for precise arrhythmia classification	Discover cardiac and non-cardiac diseases
Input Data	Raw 1D signal	2D scalogram	1D segmented	Raw 1D / masked	ECG Shape	RR	Raw 1D signal	Raw 1D signal	Raw 1D signal
Sequence Length	5000 (10 seconds)	Heartbeat-level	Segment-level	Variable (Transformer)	Beat-by-beat	Sequence of RR intervals	Beat-by-beat	Segment / Beat-by-beat	10 seconds
Model Type	1D CNN AE	2D CNN AE	LSTM autoencoder	Foundation model	Sparse AE	CAE & LSTM	Adversarial AE (TCN)	Low-Rank Attention AE	Denoising Autoencoder
Advantages	Unconstrained end-to-end learning	Time-frequency representation	Isolates P-wave (PreQ)	Captures deep temporal context	Isolates shape	Captures long-term rhythm patterns	Learns both subtle and large features	Finds relationships across the whole signal	Unsupervised disease profiling
Detection / Prediction	Detection	Detection	Detection	Detection	Detection	Prediction	Detection	Detection	Both

06 Questions?

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